MUKUL SHEREKAR

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AI-PROFICIENT BIOINFORMATICIAN

Problem-solver with strong foundation, diverse skill set and expertise in bioinformatics and AI tools. Adept at bidirectional translation between biological complexities and AI frameworks. Ability to develop pipelines for analyzing multi-omics datasets and train ML models. Interested in solving problems at intersection of biology and AI.

WORK EXPERIENCE

BIOINFORMATICS & MACHINE LEARNING ENGINEER, X10e 04/25-Current

- Curate RNAseq datasets and train ML models to predict gene regulation
- Build scripts/tools for bioinformatic data analysis
- Explain biology to ML engineer and machine learning to biologist

ORISE RESEARCH FELLOW, FOOD & DRUG ADMINISTRATION, Silver Springs, MD 07/24-12/24

Project: Addressing data imbalance in mammography datasets for training, development, and evaluation of CNN algorithms

- Engineered data preprocessing pipelines for multiple mammography datasets, integrating real and synthetic datasets to enhance CNN robustness and generalization.
- Designed and executed transfer learning experiments using ConvNeXtV1, applied image augmentation, weighted sampling, and cost-sensitive learning strategies, improving model performance on diverse datasets.
- Optimized CNN models through hyperparameter tuning, performance evaluation (ROC-AUC, confusion matrices, uncertainty analysis), and developed comprehensive documentation for reproducibility.

PROJECTS

VIRTUAL CELL CHALLENGE: Predict effects of perturbation in held out cell type 05/25-

- Translate biology (scRNAseq data) into graph embeddings to design encoders
- Experiment with architectures of published papers to improve the metrics

ECOSYSTEM: Model Context Protocol (MCP) based integrated analysis environment 05/25-

- Scalable MCP architecture enabling addition of new analysis techniques while maintaining unified agent access and context awareness
- Hybrid routing system deciding whether to route user requests to local python function through MCP servers or via LLM function calling. This approach can improve performance by bypassing LLM calls for deterministic operations
- Dynamic server discovery system that allows new analysis servers to be added, configured and loaded without any code changes to core system

SKILLS

OS: Linux (HPC) | **Programming Languages**: Python, Bash, Perl, R, Jupyter Notebooks | **Libraries**: PyTorch, Tensorflow, Biopython, Cobra, Scikit-Learn, Plotly, | **Software Engineering**: CI/CD, GitHub, Docker, SLURM, Snakemake, AWS | **Database**: MySQL, MongoDB | **Genomics**: RNA-Seq, ATAC-Seq, scRNA-seq | **Models**: DNA/ProteinBERT, AlphaFold, ProteinMPNN, RFDiffusion, ESM2 | **Bioinformatics**: BLAST, ENSEMBL, UCSC GB | **ML/DL**: Transformers, CNN, GNN, Autoencoders, GAN, Diffusion Models, LLM

Courses

Creating AI-enabled Systems| Deep Neural Networks| Applied Machine Learning| Foundation of Algorithms| Data Structures| Systems Biology| Modeling & Simulation of Complex Systems| Principles of Database Systems| Biological Databases and Tools | Molecular Biology | Epigenetics | Biochemistry

Education

- **Master of Science (MS)**, Bioinformatics, Johns Hopkins University, Baltimore, MD 2025
- **Master of Biotechnology (MBiot)**, Texas A&M University, College Station, TX 2010
- **Bachelor of Technology(B.Tech)**, Biotechnology, Vellore Institute of Technology, Vellore, India 2008